

# The impact of Bias in Healthcare

By: Luke Rudiak, Aspen Wang, Eric Zhang





# Introduction and Data



# Introduction

- The healthcare industry uses Machine Learning algorithms for prediction tasks such as estimating a patient's healthcare costs and health insurance claims.
- Studies have shown how racial bias in algorithms used to predict potential healthcare needs for patients lead to fewer resources being spent on minority group patients(Obermeyer et al, 20
- Our Project analyzes the impact biased (Racial, Gender, Age) data has on the creation of biased prediction algorithms using 3 algorithms: Linear, Polynomial, and KNN Regression

**Research Question:** How do different biases during data collection and the flexibility of a model affect algorithmic biases when predicting a patient's hospital visit billing amount?

- Findings show the algorithms generally performed worse towards group(s) experiencing biased during data collection



# Data

- Dataset is a Synthetic Healthcare dataset created by Prasad Patil on Kaggle with 15 columns and 10,000 unique synthetic patient entries
- Variables: Blood Type, Medical Condition, Insurance Provider, Medication, Results, Admission Type, Billing Amount, Doctor, Hospital, Room Number, Discharge Date, Date of Admission, Test Result
- Created Race, Admission Length (Date of Admission - Discharge Date), and Age Group variables due to potential correlation with Billing Amount, and for creating biased datasets
- Added Correlation between our predictors and outcome (Billing Amount), because the synthetic nature of the dataset made the predictors have no natural correlation



# Exploratory Data Analysis

- Limited data cleaning due to synthetic nature (no N/A, miscellaneous, and/or repeated values/rows)
- Removed irrelevant columns and converted to title case
- Manually created features we speculated had correlation with billing amount. This included:
  - Race column, which was randomly generated based off of demographic data from the US census, and used a preset seed for replicability.
  - Admission length column, generated by combining admission date and discharge date. This was the single largest factor in calculating billing amount.
  - Age group column, created by categorizing age groups into buckets. This allowed us to better examine age related bias.



# Exploratory Data Analysis

- When performing linear regression testing on our dataset as part of our initial analysis, it became clear that no correlation existed between any of our variables.
- To counteract this, we researched real-world statistics on how each feature may impact billing amount and incorporated that correlation into our data.
- The values within our predictors had to be converted into categorical predictors, or be replaced with “dummy variables”. This was achieved by utilizing the `get_dummy()` function from `pandas` to properly represent multi-level string predictors.

## Correlation between predictors and outcome Billing Amount

| Predictor         | Correlation w/ Outcome                                                                             | Adjustment in Data                                                                         | Source(s)                                                                                                                                                                                                                                                                   |
|-------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gender            | Women's Billing Amounts on average are 18% higher than that of Men.                                | If patient gender = Female, Billing Amount * 1.18                                          | <a href="https://www.weforum.org/stories/2023/10/healthcare-equality-united-states-gender-gap/">https://www.weforum.org/stories/2023/10/healthcare-equality-united-states-gender-gap/</a>                                                                                   |
| Admission Length  | Average cost per day of hospital visit is \$3,025                                                  | Find total cost of stay using Admission Length x \$3,025 and Add to current Billing Amount | <a href="https://nchstats.com/average-cost-of-hospital-stays-in-us/">https://nchstats.com/average-cost-of-hospital-stays-in-us/</a>                                                                                                                                         |
| Medical Condition | <b>Diabetes:</b> Roughly \$19,736 annual average medical expenditures(2022) translates to \$54/day | Admission Length x 54<br>Added to the current patient Billing Amount                       | <a href="https://pubmed.ncbi.nlm.nih.gov/37909353/">https://pubmed.ncbi.nlm.nih.gov/37909353/</a>                                                                                                                                                                           |
|                   | <b>Asthma:</b> Roughly \$3,266 annual per-person medical cost (2015 dollars) translates to \$9/day | Admission Length x 9<br>Added to the current patient Billing Amount                        | <a href="https://cms.illinois.gov/benefits/stateemployee/bewell/financialwellness/economic-impact-of-asthma.html?utm_source=chatgpt.com">https://cms.illinois.gov/benefits/stateemployee/bewell/financialwellness/economic-impact-of-asthma.html?utm_source=chatgpt.com</a> |

|                |                                                                                                   |                                                                      |
|----------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Admission Type | Emergency Room visits are about \$2,000 more expensive per visit compared to non-emergency visits | If patient Admission type = Emergency, add 2,000 to Billing Amount   |
| Race           | <b>Asian:</b> Baseline \$0 additional costs                                                       | No change to Billing Amount                                          |
|                | <b>Hispanic:</b> About \$1,333 more compared to baseline                                          | Multiple Billing Amount by 1.28x for patients with Race = Hispanic   |
|                | <b>Black:</b> \$2,669 more compared to baseline                                                   | Multiple Billing Amount by 1.57x for patients with Race = Black      |
|                | <b>White:</b> \$3,449 more compared to baseline                                                   | Multiple Billing Amount by 1.74x for patients with Race = White      |
|                | <b>Multi-Race:</b> \$4,584 more compared to baseline                                              | Multiple Billing Amount by 1.98x for patients with Race = Multi-Race |
| Age Group      | <b>Under 21:</b> Ages 21 and under on average spend about \$2,000 on healthcare                   | Add \$2,000 to patients billing amount where Age Group = 0-21        |
|                | <b>Age 22-40:</b> Ages 22-40 on average spend about \$3,000 on healthcare                         | Add \$3,000 to patients billing amount where Age Group = 22-40       |
|                | <b>Age 45-65:</b> Ages 45-65 on average spend about \$6,500 on healthcare                         | Add \$6,500 to patients billing amount where Age Group = 45-65       |
|                | <b>Age 65+:</b> Ages 65+ on average spend about \$11,000 on healthcare                            | Add \$11,000 to patients billing amount where Age Group = 65+        |





# Methods

-Several different algorithms were utilized on many different subsets of our data in order to provide a truly comprehensive analysis of any potential bias that might be represented.

- Linear Regression: Assumes a linear relationship between predictors, and allows you to see bias in model weights directly. This model was selected due to its simplicity, as it provides a clear understanding of how variables affect the outcome, and creates a baseline from which to compare other models.
- K-Nearest Neighbors: Predicts outcomes based on data distribution, by grouping similar data points into clusters. This is particularly effective at detecting data-driven bias, and by comparing to linear regression we can see how much of the model's bias comes from the data itself versus the algorithm's assumptions.
- Polynomial Regression: Non-linear regression that outperforms linear regression at predicting complex relationships. Allows us to select the degree for the polynomial, which means we can get the best possible fit for the data.



## Methods

-Model training was performed on our original dataset, which matched real-world distributions of Age, Race, and Gender statistics.

-Next, nine biased subsets of our data were built upon unequal splits of these demographics. These consisted of:

| Gender   | Race      | Age           |
|----------|-----------|---------------|
| 70% Male | 70% White | 70% aged 0-40 |
| 80% Male | 80% White | 80% aged 0-40 |
| 90% Male | 90% White | 90% aged 0-40 |



## Methods

-Each subset was constructed by randomly sampling participants from the original dataset, until we had groupings that match our desired ratios exactly. Each of the three predictive algorithms were run on each of the subsets, measuring the  $R^2$  value, the RMSE, and predicted billing amount/standard deviation for each split.

-By utilizing this split-dataset approach, we were able to:

- Perform controlled experiments on demographic imbalance.
- Examine how models behave as the underlying population structure changes.
- Extrapolate which characteristics are most sensitive.
- Separate data driven bias from algorithm driven bias.



# **Analysis & Findings**



# Aggregated Model Performance

- **Polynomial:** Most Accurate
- **Linear Regression**
- **KNN:** Least Accurate

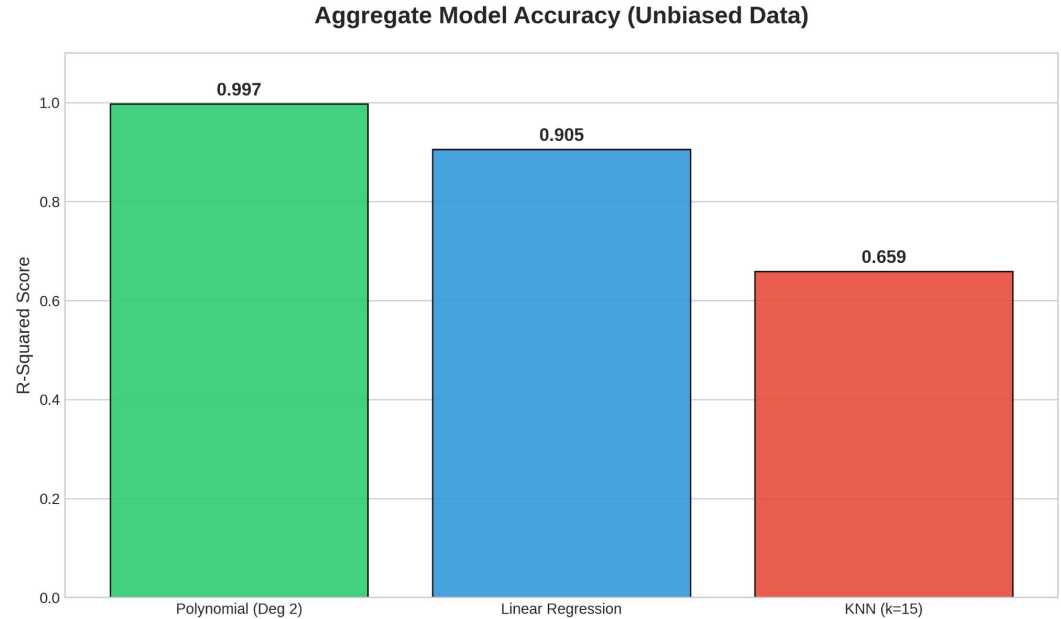


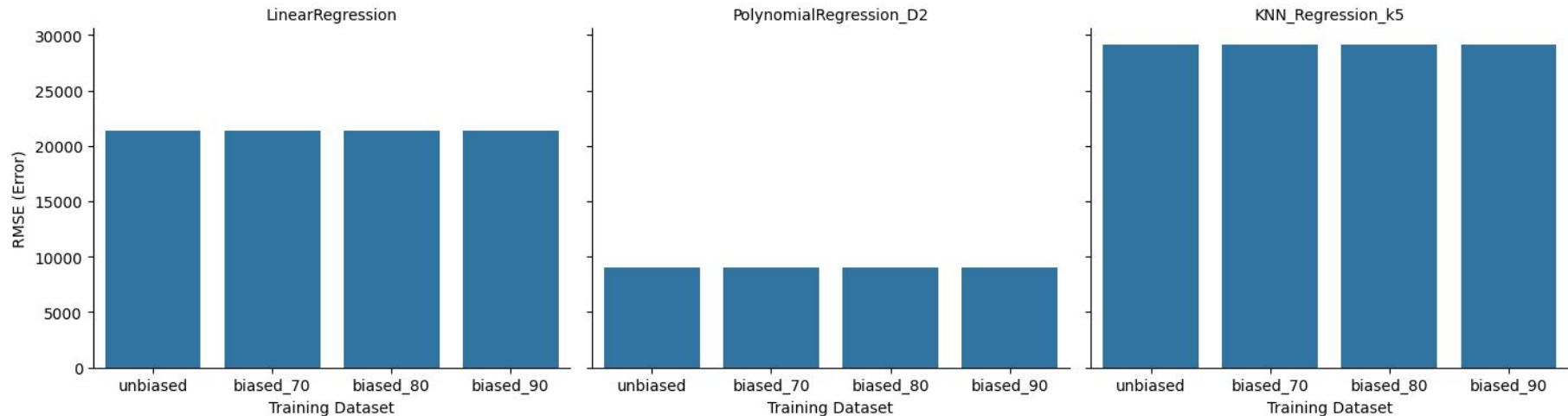
Fig: Model Performance (R-Square)



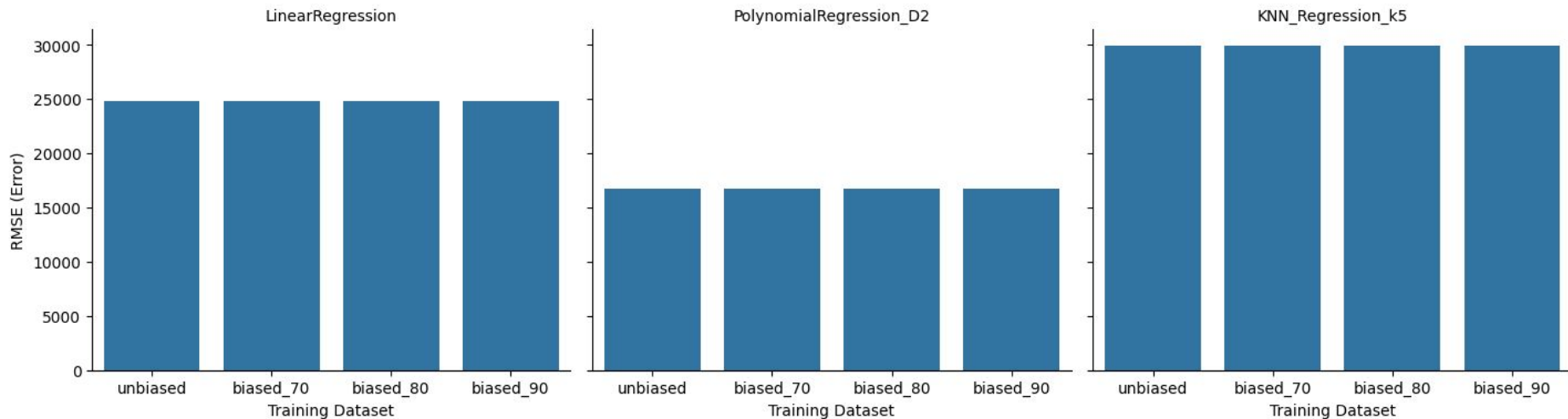
# Impact of Biased Datasets on Aggregate Performance

- We initially assumed that the biased datasets would result in severe aggregated biases
- **Contrary to expectation**, introducing severe data collection biases **did not** reduce aggregate model accuracy or propagate bias into predictions.
- Our findings suggest that the choice of models are the primary factors of the performance disparities.

Model Performance on the UNBIASED Test Set



Model Performance on the BIASED 90% Test Set





## Impact of biased datasets

- Age Group Bias
- Racial Bias
- Gender Bias



Group RMSE





# Impact of biased datasets

## **Polynomial Regression (Most Accurate):**

- Achieved the best overall accuracy ( $\text{MSE} \approx 1.34 \times 10^7$ ).
- Fairness Issue: High disparity in error rates. Error for White patients ( $\text{RMSE} \approx 10,742$ ) was  $2.5\times$  higher than for African American patients ( $\text{RMSE} \approx 4,397$ ).

## **KNN Regression (Worst Fairness):**

- Highly sensitive to sample size deficiencies.
- Extremely inaccurate for underrepresented groups (Multiracial/Asian).
- Multiracial RMSE:  $\approx 50,000$  (approx.  $2.5\times$  larger than the model's best-performing group).

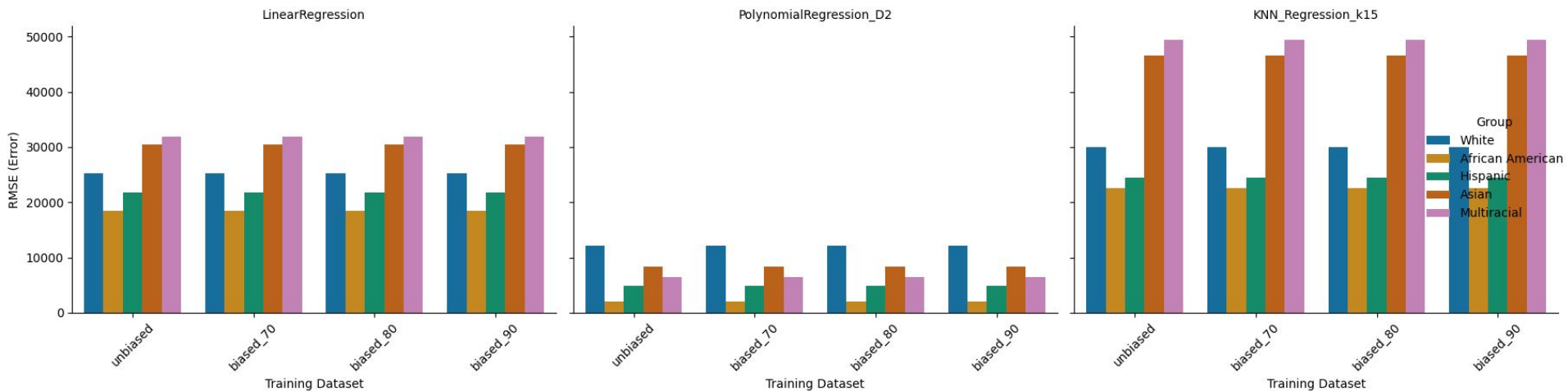
## **Linear Regression (Most Consistent):**

- Shown notable disparities (African American  $\text{RMSE} \approx 18,900$  vs. Multiracial  $\approx 27,000$ ).
- Result: Demonstrated the smallest relative variability between best and worst-performing groups compared to Poly and KNN.

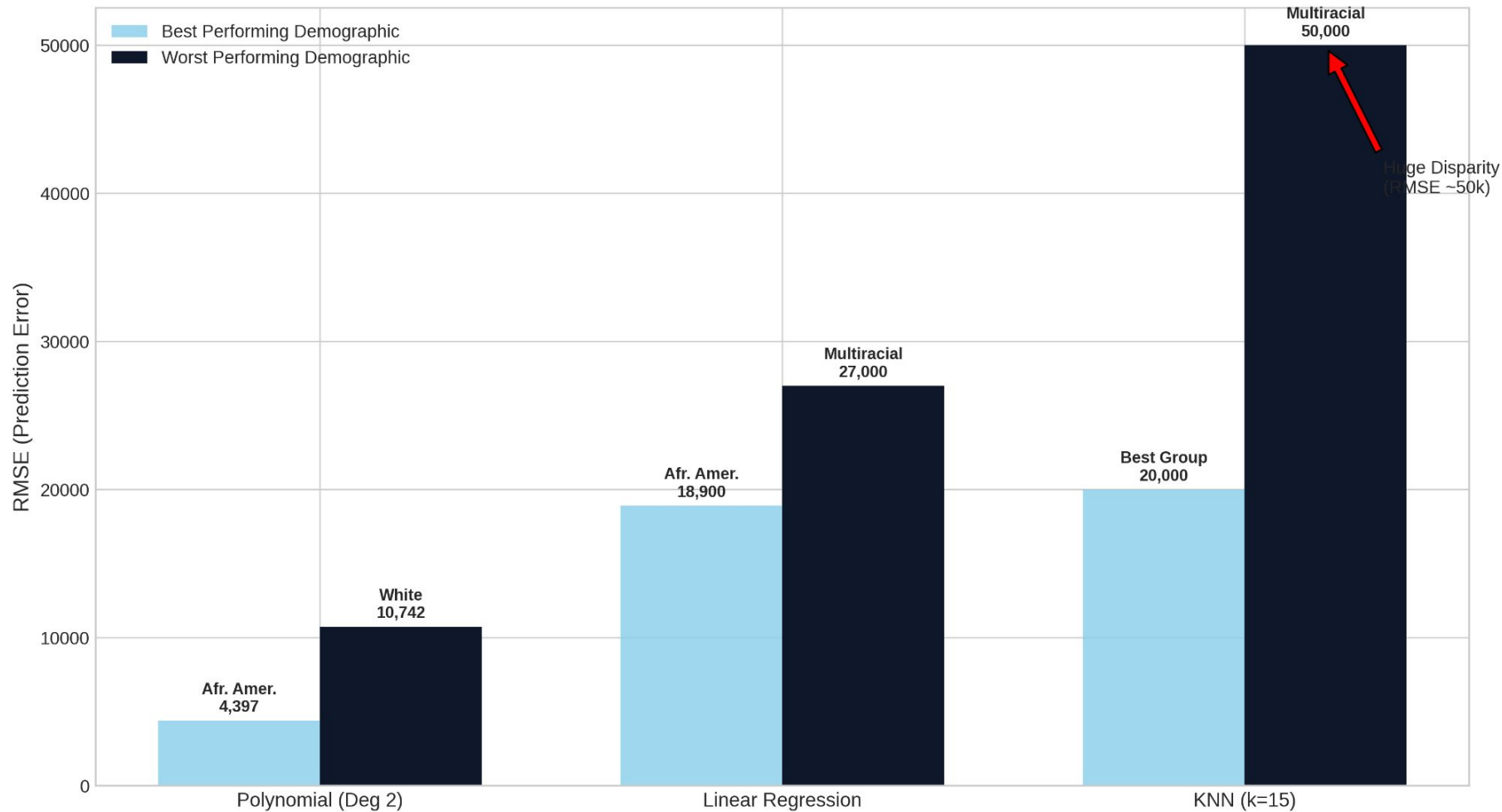


# Analysis & Findings

Group Bias Analysis



# Algorithmic Fairness: Disparity Between Groups





## Summary

- While bias **did not** propagate from the **data splits**, significant **inherent bias** exists within the **models** themselves.
- **Model flexibility**, rather than data collection bias, was the primary driver of algorithmic performance disparities.
- A model that performs best in overall accuracy may not be the most appropriate choice when fairness and equitable outcomes are our primary concerns.



# **Discussion + Conclusion + Future Work**





# Summary of Findings

## No Propagated Bias

Contrary to intuition, training on extremely skewed datasets (e.g., 90% White) did not degrade aggregate accuracy or propagate bias into predictions.

Demographic under-representation alone was not enough to harm the predictive performance of these models.

## Inherent Bias & Flexibility

Significant bias originated from the model choice, not the data.

High flexibility models (KNN) produced the most disparate error rates, while rigid models (Linear Regression) offered the most equitable (though less accurate) outcomes.

## Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification\*

Joy Buolamwini

MIT Media Lab 75 Amherst St. Cambridge, MA 02139

JOYAB@MIT.EDU

Timnit Gebru

Microsoft Research 641 Avenue of the Americas, New York, NY 10011

TIMNIT.GEBRU@MICROSOFT.COM

Editors: Sorelle A. Friedler and Christo Wilson

### Abstract

Recent studies demonstrate that machine learning algorithms can discriminate based on classes like race and gender. In this work, we present an approach to evaluate bias present in automated facial analysis algorithms and datasets with respect to phenotypic subgroups. Using the dermatologist approved Fitzpatrick Skin Type classification system, we characterize the gender and skin type distribution of two facial analysis benchmarks, LJB-A and Adience. We find that these datasets are overwhelmingly composed of lighter-skinned subjects (79.6% for LJB-A and 86.2% for Adience) and introduce a new facial analysis dataset which is balanced by gender and skin type. We evaluate 3 commercial gender classification systems using our dataset and show that darker-skinned females are the most misclassified group (with error rates of up to 34.7%). The maximum error rate for lighter-skinned males is 0.8%. The substantial disparities in the accuracy of classifying darker females, lighter females, darker males, and lighter males in gender classification systems require urgent attention if commercial companies are to build genuinely fair, transparent and accountable facial analysis algorithms.

**Keywords:** Computer Vision, Algorithmic Audit, Gender Classification

### 1. Introduction

Artificial Intelligence (AI) is rapidly infiltrating every aspect of society. From helping determine

who is hired, fired, granted a loan, or how long an individual spends in prison, decisions that have traditionally been performed by humans are rapidly made by algorithms (O’Neil, 2017; Citron and Pasquale, 2014). Even AI-based technologies that are not specifically trained to perform high-stakes tasks (such as determining how long someone spends in prison) can be used in a pipeline that performs such tasks. For example, while face recognition software by itself should not be trained to determine the fate of an individual in the criminal justice system, it is very likely that such software is used to identify suspects. Thus, an error in the output of a face recognition algorithm used as input for other tasks can have serious consequences. For example, someone could be wrongfully accused of a crime based on erroneous but confident misidentification of the perpetrator from security video footage analysis.

Many AI systems, e.g. face recognition tools, rely on machine learning algorithms that are trained with labeled data. It has recently been shown that algorithms trained with biased data have resulted in algorithmic discrimination (Bolukbasi et al., 2016; Caliskan et al., 2017). Bolukbasi et al. even showed that the popular word embedding space, Word2Vec, encodes societal gender biases. The authors used Word2Vec to train an analogy generator that fills in missing words in analogies. The analogy man is to computer programmer as woman is to “X” was completed with “homemaker”, conforming to the stereotype that programming is associated with men and homemaking with women. The biases in Word2Vec are thus likely to be propagated throughout any system that uses this embedding.

\* Download our gender and skin type balanced PPP dataset at [gendershades.org](https://gendershades.org)

# Connection to Literature

## *Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification*

Buolamwini & Gebru (2018) found that **representational disparities** in facial recognition led to catastrophic failure for minority groups.

Our Divergence: In healthcare billing prediction, **under-sampling did not** inherently cause **higher error rates**.

Key Insight: The harm of under-sampling becomes severe only when combined with highly flexible models (like KNN) that overfit to majority clusters.



# Limitations and Future Work

## **Synthetic Data**

While we modeled demographic distributions on real-world data, synthetic datasets cannot perfectly recreate the precision and noise of actual patient records.

## **Model Selection**

We limited our scope to three regression types. Future work should incorporate Random Forests (low overfitting) and Neural Networks (complex interactions).

## **Accuracy of Correlations**

We performed surface level research(i.e browsing the internet) to find potential correlations between our predictors and outcome. Future work includes performing in depth research or experiments to find potential trends.





# Ethical and Societal Implications

## **Accountability & Privacy**

Using synthetic data reduces patient privacy risks and clearly designates accountability for biases to the creators (us). However, this adds an extra layer of responsibility for the creator to remain unbiased during the data creation process.

## **Risk of Stereotyping**

Introducing artificial correlations without clinical backed research and or experimentation risks reinforcing negative stereotypes, potentially suggesting wrongful discrimination.

## **Conclusion:**

This work serves as a validation tool for hypotheses, not a replacement for real-world analysis. It warns against the risks of manipulating data to “discover” correlations in data.



# Q & A



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**Thank you!**